

of utility (“Lord make me chaste, but not yet”).⁴⁴ Fortunately, some economists have also exploited behavioral biases to entice people to save.

Thaler and Benartzi (2004) created the Save More Tomorrow program, described in Thaler and Sunstein’s (2009) bestseller, *Nudge*. Thaler and Benartzi exploit human nature to get people to commit to saving future wage increases. People won’t save out of today’s funds, but they will save out of funds that they haven’t yet received. When they receive those funds, they can’t be bothered to change their original plans and inadvertently end up saving. Inertia is thus enlisted on behalf of thrift at no cost to current consumption. Thaler and Benartzi found savings rates increased from 3% to over 13% when employees committed to a program where they saved more tomorrow.

Conversely, Scholz, Seshadri and Khitatrakun (2006) provocatively argue that most households have adequately saved for retirement. There is, however, no mortality risk in their model for working people, no out-of-pocket health-care risk prior to retirement, and no risky assets. It’s likely that taking into account these risks, households are not saving enough. Also, their study was completed before the Great Recession of 2008–2009, which decimated many households’ balance sheets.

The lack of savings is also reflected in the underfunding of defined benefit pension plans. This is behind Rhode Island’s travails and Raimondo’s desire to reform its pension system. In these plans, employers save on behalf of workers, but employers have not been contributing enough. Or the benefits promised are too high. Or both. In the long run, there is an immutable *adding up constraint*: the money that we spend during retirement cannot exceed the savings (plus investment earnings) that we have set aside to fund retirement.

If we don’t save enough, then we’ll have to consume less during retirement. Skinner (2007) argues that we can consume less during retirement and still be happy. There is a lot more leisure during retirement. We may place great value on this unexpected richness of time, which can offset our lower consumption. Skinner had better be right, because most people are going to need to consume a lot less after they retire.

4.3. THE FOUR PERCENT RULE

The *4% rule* was created by William Bengen (1994), a financial planner.⁴⁵ Bengen found that an investor holding 50% stocks and 50% bonds could make her portfolio last at least thirty years if she spent approximately 4% per year. In Bengen’s original formulation, the investor started retirement by consuming 4% of her wealth at that time—so she starts with a nest egg of twenty-five times her desired

⁴⁴ See Laibson (1997).

⁴⁵ The rule has many variants, several due to follow-up work by Bengen. Milevksy and Huang (2011) develop versions with mortality risk, which Bengen (1994) omits.